

Data Science Midterm Project

Analysis of Global COVID-19 Datasets

P.M.H

CS320/CS110: Fall 2025

Summary

This midterm project requires you to perform a comprehensive analysis of six provided COVID-19 datasets. You will be responsible for the entire data science lifecycle: formulating a hypothesis, cleaning and processing the data, performing exploratory data analysis (EDA), testing your hypothesis, and presenting your findings in a formal technical report. This project will test your skills in data manipulation, statistical analysis, visualization, and scientific communication.

Goals

The goals of this assignment are:

1. To develop and test a meaningful hypothesis using real-world data.
2. To gain practical experience in cleaning, merging, and managing multiple, complex datasets.
3. To apply exploratory data analysis and visualization techniques to uncover insights and patterns.
4. To practice communicating analytical findings clearly and professionally through a structured technical report.
5. To enhance critical thinking about data limitations and the implications of your results.

Before You Begin: Individual Preparation

Review Key Concepts

- **Python Libraries:** Refresh your knowledge of Pandas (for data manipulation), NumPy (for numerical operations), and Matplotlib/Seaborn (for data visualization).
- **Statistical Concepts:** Review correlation, regression, and common statistical metrics like Case Fatality Rate (CFR) and prevalence.
- **Data Cleaning:** Revisit techniques for handling missing values, standardizing data formats, and correcting data types.

Familiarize Yourself with the Datasets

You are provided with six CSV files. Understand the scope and content of each:

- `full_grouped.csv`: Detailed daily, country, and province-level data.
- `covid_19_clean_complete.csv`: Cleaned daily, country-level data.
- `country_wise_latest.csv`: A snapshot of the latest totals for each country.
- `day_wise.csv`: Aggregated global daily data without country-level detail.
- `usa_county_wise.csv`: Granular daily data for U.S. counties.
- `worldometer_data.csv`: Rich country-level demographic and testing data (e.g., population, testing rates).
- **Bonus (10pts):** COVID-19 World Vaccination Progress

Prepare Your Development Environment

- Set up your Python/Jupyter Notebook development environment (e.g., Anaconda, VS Code, Google Colab).
- Ensure all necessary libraries (pandas, numpy, matplotlib, seaborn, etc.) are installed.

Project Workflow: The Analysis Process

Phase 1: Hypothesis Formulation & Data Wrangling

- **Formulate a Hypothesis:** Based on the available data, develop a clear, testable hypothesis. For example: `"Countries with higher GDP per capita experienced a lower overall case fatality rate due to better healthcare infrastructure."` or `"Higher population density is positively correlated with a faster initial spread of the virus (cases per million) in the first 90 days."` Your hypothesis must require data from at least two different CSV files to be tested.
- **Data Cleaning and Integration:** Load all datasets. Perform necessary cleaning (handling NaNs, standardizing country names). Merge datasets as needed to create a master DataFrame for your analysis.

Phase 2: Analysis and Visualization

- **Exploratory Data Analysis (EDA):** Generate descriptive statistics. Create meaningful visualizations (e.g., time-series plots of global cases, choropleth maps of confirmed deaths, scatter plots for correlations) to explore the data.
- **Hypothesis Testing:** Perform the specific analysis required to prove or disprove your hypothesis. This could involve calculating correlation coefficients, running a simple regression, or comparing distributions between groups of countries.

After Analysis: Pair Lab Report

Overview: This lab report should follow the basic structure of a scientific or technical paper. Below, you'll find guidance on how to write each section of your report. Be sure to explain your findings clearly, provide enough detail for others to understand your methods and results, and critically discuss your approach and its limitations. You are ***required*** to use **Overleaf** for writing your paper to ensure a professional presentation.

The basic parts of a scientific or technical paper are as follows:

1. **Title and Author Information:**

2. **Abstract:** A concise summary of your project, including the hypothesis, key methods, main results, and conclusion.

3. **Introduction:** Introduce the topic (the COVID-19 pandemic), state your specific hypothesis, and explain why it is an interesting question to investigate.

4. **Data and Methodology (replaces Code Explanation):** Describe the datasets used. Detail your data cleaning, preprocessing, and integration steps. Explain the analytical methods and statistical tests you employed.

5. **Results:** Present your key findings objectively, without interpretation. Use tables and figures (e.g., plots, charts) to illustrate the results. Ensure each figure has a descriptive caption.

6. **Discussion:** Interpret your results in the context of your hypothesis. Did the evidence support or refute it?

- ***Note:*** If you encountered issues (e.g., messy data, analytical challenges), describe what went wrong and how you addressed it.
- This section should provide a critical reflection on the limitations of your analysis and the data itself. What are the potential biases or confounding variables?
- It should also demonstrate your ability to identify problems and propose solutions or directions for future research.

7. **Conclusion:** Briefly summarize the project and your main conclusions. Reiterate whether your hypothesis was supported.

8. **References:** Cite any external sources or libraries you used.

9. **Appendices:**

- ***Note:*** Include the full source code of your program (e.g., Python script or Jupyter Notebook export) in the appendices. The code should be neatly formatted and properly commented to enhance readability.
- Using Overleaf is ***required*** for writing your report and formatting the code, as it ensures consistency and professional presentation.

Important Notes:

- Pay attention to formatting and clarity. Your report should be well-organized, easy to follow, and free of grammar or spelling errors.
- I am ***Looking for*** critical thinking and understanding in the report.

Academic Integrity Policy

- **Cheating is strictly prohibited.** Copying directly from AI-generated content or any other published work without significant editing or proper citation will not be tolerated.
- Any instance of plagiarism or academic dishonesty will result in a score of zero for the assignment or project, and a report will be made to the school administration.
- Students are expected to submit their own original work. If you need help or clarification, seek assistance in a legitimate manner, and always ensure proper citation if you reference external sources.
- Remember, integrity is essential to your learning process and academic growth. Abiding by the rules helps maintain a fair and respectful academic environment for everyone.

Grading Rubric

Component	Expectations	Weight
Code & Data Processing	Clear, efficient, and well-commented code. Correctly loads, cleans, and merges data from multiple sources. Proper error handling demonstrated. Attach code to Appendix.	25%
Analysis & Results	Appropriate use of EDA and visualization techniques. Correct implementation of statistical methods to test the hypothesis. Results are presented clearly and accurately in figures and tables.	30%
Discussion & Critical Thinking	Insightful interpretation of results. Thorough discussion of the findings in relation to the hypothesis. Acknowledges limitations of the data and methodology. Demonstrates critical thinking.	25%
Report Quality & Writing Style	Professionally formatted report. Well-organized, clear writing with proper grammar and punctuation. Logical flow between sections and concise explanations. All required sections are present.	20%

Note: Points will be deducted for poor writing quality (e.g., grammar, clarity, or structure issues) that affect the overall readability of the paper.

Upload to Canvas the following:

- **PDF report:** LastName_FirstName_Midterm_CSXXX.pdf
- **LaTeX code:** LastName_FirstName_Midterm_CSXXX.tex
- **Code File:** LastName_FirstName_Midterm_CSXXX.ipynb (or .py)